

Custom Corrective Retrieval-Augmented Generation (CRAG-CAP)

Master Technical Report: Formulas, Mathematical Calculations, Data Tables & Worked Examples

Phase 1: Problem Statement & CRAG 3-Way Core Concept

Standard RAG blindly trusts retrieved text. When retrieval brings noisy or incorrect context, LLMs hallucinate. CRAG introduces a 3-Way Decision Controller evaluating passage relevance before generation:

Decision Pathway	Condition / Rule	Operational Execution Logic
CORRECT	MaxScore >= +0.5920	High confidence. Refine local text; SKIP web search.
AMBIGUOUS	-0.9950 <= MaxScore < +0.5920	Uncertain confidence. Combine refined local text + Serper Google Search.
INCORRECT	MaxScore < -0.9950	Pure noise. Discard local text; execute query rewriting & Serper Google Search.

Phase 2: Evaluator Model Training & Classification Performance

We fine-tuned **google-t5/t5-small (60.5M parameters)** on PopQA training triples. Evaluator classification metrics:

Metric Name	Mathematical Formula	Evaluator Tested Result
Classification Accuracy	$Acc = (TP + TN) / (TP + TN + FP + FN)$	94.40%
Relevance Precision	$Precision = TP / (TP + FP)$	92.80%
Relevance Recall	$Recall = TP / (TP + FN)$	95.60%
Relevance F1-Score	$F1 = 2 * (Precision * Recall) / (Precision + Recall)$	94.18%

Phase 3: Mathematical Score Calculation & Score Adapter Derivation

At decoder step 0, T5 outputs raw logits L1 (token '1', ID 209) and L0 (token '0', ID 3). Softmax normalizes them into relevance probability P(1):

Probability Normalization: $P(1) = \exp(L1) / [\exp(L1) + \exp(L0)]$

Linear Score Adapter Derivation: $S(P) = a * P(1) + b$. Solving boundary conditions $S(0) = -1.0$ and $S(1) = +1.0$ yields:

CRAG_Score = 2.0 * P(1) - 1.0 in range [-1.0, +1.0]

Worked Numerical Calculation Example:

- Raw T5 Logits: L1 = +4.50, L0 = +1.20
- Exponentials: $\exp(4.50) = 90.017$, $\exp(1.20) = 3.320$
- Relevance Probability: $P(1) = 90.017 / (90.017 + 3.320) = 90.017 / 93.337 = \mathbf{0.9644 (96.44\%)}$
- CRAG Score Adapter: $Score = 2.0 * 0.9644 - 1.0 = \mathbf{+0.9288}$ (Triggers **CORRECT** path since $+0.9288 \geq +0.5920$).

Phase 4: 3-Way Threshold Probability Reverse Mapping Math

Reverse-mapping scores back to probabilities via $P = (S + 1) / 2$:

- Upper Threshold (gamma1 = +0.5920) ==> $P(1) = (+0.5920 + 1) / 2 = 79.60\%$ Probability (High Confidence)**
- Lower Cutoff (gamma2 = -0.9950) ==> $P(1) = (-0.9950 + 1) / 2 = 0.25\%$ Probability (Pure Noise Cutoff)**

Phase 5 & 6: Knowledge Processing & Infrastructure Integration

- **Passage Decomposition:** Paragraphs split into sentence sub-strips (selection mode, ~1-3 sentences).
- **Sub-Strip Sentence Filtering:** Evaluator deletes any sub-strip with Score < -0.9950.
- **Search Query Rewriting:** Groq 120B LLM rewrites question: *"What is Henry Feilden's occupation?"* ==> *"Henry Feilden occupation"*.
- **Web Search API:** Real Google Serper API (Serper.dev) fetches live organic Google search snippets.
- **Multi-Key Auto-Rotation:** 4 Groq API keys (800,000 TPD) with HTTP 429 automatic retry backoff.

Phase 7: Full 1,385-Query Benchmark Results & Metric Formulas

Metric Name	Mathematical Formula	Tested Result (N = 1,385)
Exact Match Accuracy (EM)	$EM = (\text{Count of Exact Substring Matches}) / N * 100$	58.63% (812 / 1,385)
QA Token Recall	$Recall = \text{Tokens}(\text{Pred}) \cap \text{Tokens}(\text{Gold}) / \text{Tokens}(\text{Gold}) $	59.25%
QA Token Precision	$Precision = \text{Tokens}(\text{Pred}) \cap \text{Tokens}(\text{Gold}) / \text{Tokens}(\text{Pred}) $	11.65%
QA Token F1-Score	$F1 = 2 * (Precision * Recall) / (Precision + Recall)$	18.87%
ROUGE-L F1-Score	LCS-based F1 Similarity	18.87%

Pathway Metrics Breakdown (N = 1,385 Queries):

Pathway	Queries	% Dataset	EM Accuracy	Token Recall	Token F1
CORRECT	583	42.09%	70.50%	71.90%	22.66%
AMBIGUOUS	661	47.73%	48.41%	48.55%	15.38%
INCORRECT	141	10.18%	57.45%	57.09%	19.59%
TOTAL SYSTEM	1,385	100.00%	58.63%	59.25%	18.87%

Benchmark Performance Comparison Table:

System Architecture	Evaluator Model	Generator LLM	PopQA EM Accuracy
Standard RAG Baseline	None	LLaMA-2 7B	44.80%
Original Published CRAG (Yan 2024)	T5-Large (770M)	Self-RAG 7B	53.90%
OUR CUSTOM CRAG IMPLEMENTATION	T5-Small (60.5M)	Groq 120B	58.63% (+4.73% over paper)